
Assignment 2 - Swedish Language Tutor Agent

Retrieval-Augmented Beginner Swedish Practice with Memory and Agent Tools

Project Links

GitHub: <https://github.com/SeanSha/UU-SwedishLanguageTutorAgent>

Dataset: <https://huggingface.co/datasets/SeanSha30/swedish-pre-a1-learning-agent-dataset>

Demo Video: https://drive.google.com/file/d/1K_oObTMqPvRDLnkd9IgnKK3cNIyfCe53/view?usp=sharing

1 Introduction

This project builds a small Swedish language tutor agent for absolute beginner / pre-A1 learners. It helps learners practise everyday Swedish scenarios, such as food shopping, transport, health, family, home, and social introductions. Instead of working as a general chatbot, the system grounds each lesson in retrieved teaching material, including dialogue plans, vocabulary, and sentence examples.

The project connects AI agents with Information Retrieval through tool-based actions such as lesson retrieval, dialogue generation, quiz generation, answer checking, memory update, and text-to-speech. The aim is to show how retrieval, memory, and learning actions can work together in a practical language-learning prototype.

2 System Design

The application is implemented as a Python Gradio web app. The interface includes topic selection, a lesson guide, quiz interaction, audio buttons, and a memory review quiz. The central class is `SwedishSpeakingAgent` in `agent.py`, which coordinates retrieval, LLM generation, quiz feedback, text-to-speech, and learner memory. The simplified learning flow is shown in Figure 1. A learner chooses a topic, the system retrieves relevant teaching material, and the agent turns it into guided dialogue, quizzes, translation support, and audio practice. Mistakes are saved as memory signals for later review.

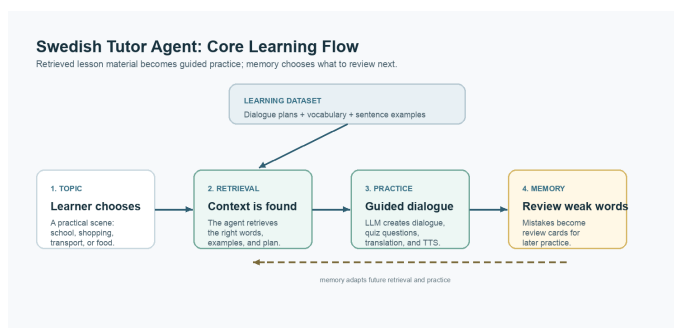


Figure 1: Core learning flow of the Swedish tutor agent.

The internal structure is shown in Figure 2. Retrieval, dialogue control, quiz generation, feedback, text-to-speech, memory, and LLM connection are separated into tools. This keeps the prototype simple to demonstrate while still making it extendable.

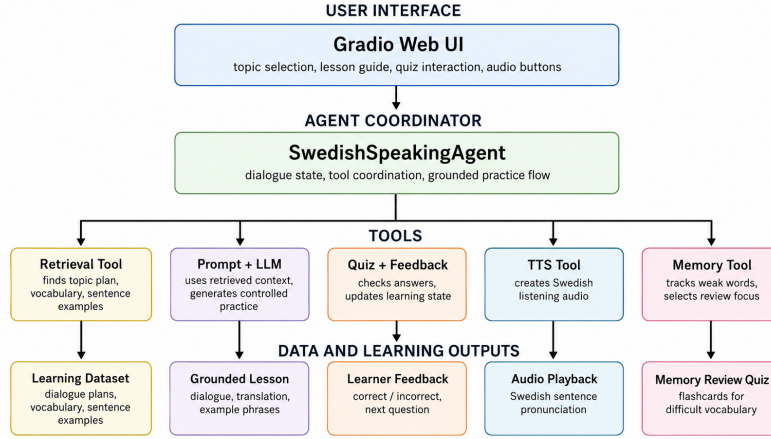


Figure 2: Internal structure of the tutor agent. The `SwedishSpeakingAgent` coordinates the interface, retrieval, LLM client, quiz feedback, TTS, memory, and review quiz.

The app supports both local and online modes. Local mode connects to an Ollama server and uses a locally installed model. Online mode uses an OpenAI-compatible chat completion client, with the API key provided through the interface or `.env` settings.

3 Information Retrieval Component

The core IR component is a structured retrieval pipeline based on three local cached files: `data/dialogue_plans.jsonl`, `data/sentence_examples.jsonl`, and `data/vocabulary.jsonl`. These files come from the public Hugging Face dataset and provide the tutor’s teaching material.

The retrieval process starts from the learner’s selected topic. The system retrieves a matching dialogue plan, related Swedish sentence examples, and topic vocabulary. The plan gives the lesson structure, the examples provide concrete language input, and the vocabulary supports explanations, quizzes, and memory review.

The retrieved material is assembled into the LLM prompt context, so the agent does not generate dialogue freely from general knowledge. Its output is guided by the selected topic, retrieved lesson plan, sentence examples, vocabulary data, and difficult words stored in learner memory.

4 Memory and Agent Actions

The agent stores a simple local memory in `memory/user_memory.json` at runtime. This file is ignored by Git, so personal learning progress is not uploaded. Incorrect answers increase the review weight of related words, while correct answers gradually reduce it.

Memory affects later practice by prioritising difficult words in retrieval, quizzes, and a separate Memory Review Quiz. This turns remembered learner behaviour into follow-up practice, rather than only showing previous statistics.

5 Conclusion and Future Work

This project shows how current language technologies can be combined to build a small language-learning app. However, the exploration also suggests that good learning content cannot be produced by engineering alone. Experienced educators are essential to design suitable learning goals, scenarios, and progression, while engineers can help transform this material into structured prompt context and interactive practice flows. In this sense, such tools should be seen as support for teachers and learners, helping students practise in a more convenient and engaging way.

From a technical perspective, several directions could be explored further. Future versions could add automatic speech recognition, more natural Swedish text-to-speech, and stronger support for spoken interaction. Another possible direction is to connect the tutor to a virtual avatar that can produce natural lip movements, facial expressions, and body gestures during dialogue practice. In the long term, this type of system could also be deployed on a social robot to support more natural human–computer interaction in language learning.